

Class 32 2D Maps

- There is a weekly project update due on Friday
- Your progress report slides are due Monday.
- The progress report presentations will be Monday and Wednesday

Big picture

We have seen a strange attractor and chaos in a 3D continuous time dynamical system.

Prof Ghrist proved the presence of chaos in 1d (non-invertible) maps that have stretching and tearing or stretching and folding. One of these was a model for the Lorenz system.

Today we will look at a 2D map that provides a model of chaos and will look a little more at the geometry of the period-doubling cascade.

How to follow up on this class: AM 105/109, Math 118r, AM 91r, MIT courses (search 'chaos' for a couple of options that overlap with our course but go beyond it)

Questions

1. (Our first 2D map)

The Baker's map is given by

$$B(x_n, y_n) = (x_{n+1}, y_{n+1}) = \begin{cases} (2x_n, ay_n) & \text{for } 0 \leq x_n \leq \frac{1}{2} \\ (2x_n - 1, ay_n + \frac{1}{2}) & \text{for } \frac{1}{2} \leq x_n \leq 1 \end{cases}.$$

It is illustrated by Figure 12.1.4 of the text, shown below.

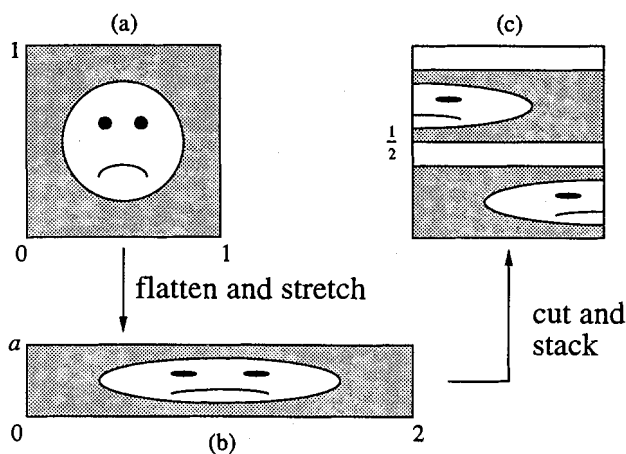


Figure 12.1.4

The Baker's transformation is a simple model with chaotic dynamics. We can reason about the long term behavior under this map by thinking geometrically, or by rewriting the map as a shift map on a sequence of numbers.

(a) The Baker's transformation is the first invertible map that we are working with.

- Explain why the $2x \bmod 1$ map, the $2x$ tent map, and the logistic map are each not invertible.
- For the Lorenz equations and the Rossler system, you can find a backwards time version of the system (the systems are time reversible). An invertible map captures this time reversability better than a non-invertible map. Why is that?

- (b) $B(x_n, y_n)$ is equivalent to the procedure of stretching by 2, flattening by a , then cutting and stacking, that is shown in the figure. Convince yourself and your team that this is the case. For $a < 1/2$ there are blank horizontal spaces present after stacking. These are regions of the unit square that nothing in the original domain maps to.
- (c) Sketch what will happen after one more iterate of the map shown in the figure. Include the face and the bands of empty space (white space on this sheet).
- (d) This process might remind you of forming the Cantor set. Sketch some representation of the limiting set. How would you describe this set?
- (e) Explain why we can't use similarity dimension to determine the dimension of the limiting set. Recall that the similarity dimension was found by scaling down the original object and making a fixed number of copies of it.

The box dimension is another way to compute fractal dimension. If you are interested, you can read about it in our course textbook.

2. (a) (12.1.5) Much of the analysis people do on maps is done via *symbolic dynamics*.

For the area preserving Baker's map, consider a binary representation of a point in the unit square:

$$(x, y)_2 = (0.a_1a_2a_3..., 0.b_1b_2b_3...)$$

where $a_1 = 0$ indicates the point has $0 \leq x < \frac{1}{2}$ and $a_1 = 1$ indicates the point has $\frac{1}{2} \leq x < 1$. Given the binary representation of (x, y) , find the binary representation of $B(x, y)$.

Multiplying a coordinate by 2 has the effect of shifting the decimal place once to the right.

- (b) Represent the point (x, y) as $...b_3b_2b_1.a_1a_2a_3...$. In this notation, what is $B(x, y)$?
- (c) Use the binary version of the map to show that B has a single period-2 orbit. Plot the locations of the two points involved in the orbit in the unit square.

1a: $f(2) = f(1)f(1) = a^2$.

b: $f(n) = f((n-1)+1) = f(n-1)f(1) = af(n-1) = a^n$.

c: $f(1) = f(1/2 + 1/2) = f(1/2)^2$ so $f(1/2) = \sqrt{a}$.

d: $f(p/q) = f(1/q + 1/q + \dots + 1/q) = f(1/q)^p$. $f(q/q) = f(1/q)^q = a$ so $f(p/q) = a^{p/q}$.

2. a: $g(0) = 1$. $g(0) = \alpha g(g(0)) = 1 \Rightarrow \alpha = 1/g(1)$.

b: $g(x) = \alpha g(g(x/\alpha))$. $h(x) = \mu g(x/\mu)$. Does $h(x) = \alpha h(h(x/\alpha))$? $h(x) = \mu g(x/\mu) = \mu \alpha g(g(x/(\alpha\mu))) = \alpha \mu g(\mu g(x/\alpha\mu)/\mu) = \alpha h(\mu g(x/\alpha\mu)) = \alpha h(h(x/\alpha))$ so h does satisfy it.

c: x^* has $g(x^*) = x^*$. $g(\alpha x^*) =$

3. See previous handout

4. a: The x coordinate is right shifted by the stretch (the map multiplies it by 2), so it becomes $a_1.a_2a_3a_4...$. Cutting and stacking turns it into $0.a_2a_3a_4...$. For the y coordinate, it depends on the x coordinate. If $a_1 = 0$ then y becomes $0.0b_1b_2...$ while if $a_1 = 1$ then y becomes $0.1b_1b_2...$. So $(0.a_1a_2a_3, 0.b_1b_2b_3) \mapsto (0.a_2a_3..., 0.a_1b_1b_2...)$.

b: $...b_3b_2b_1.a_1a_2a_3... \mapsto ...b_2b_1a_1.a_2a_3a_4...$ so the map acts as a shift map on this representation.

c: For a period-2 orbit, we are looking for a binary number that returns to itself after two shifts. These are the repeating fractions $...101010.101010...$ and $...010101.010101...$. Their coordinates are given by $x = \frac{1}{2} + \frac{1}{8} + \dots$, $y = \frac{1}{4} + \frac{1}{16} + \dots$ and vice versa. Thus $x - \frac{1}{4}x = \frac{1}{2} \Rightarrow x_1 = \frac{2}{3}$ and $y - \frac{1}{4}y = \frac{1}{4} \Rightarrow y = \frac{1}{3}$. The points are $(\frac{2}{3}, \frac{1}{3})$ and $(\frac{1}{3}, \frac{2}{3})$.

2D (invertible) map models

Baker's transformation (stretch and tear):

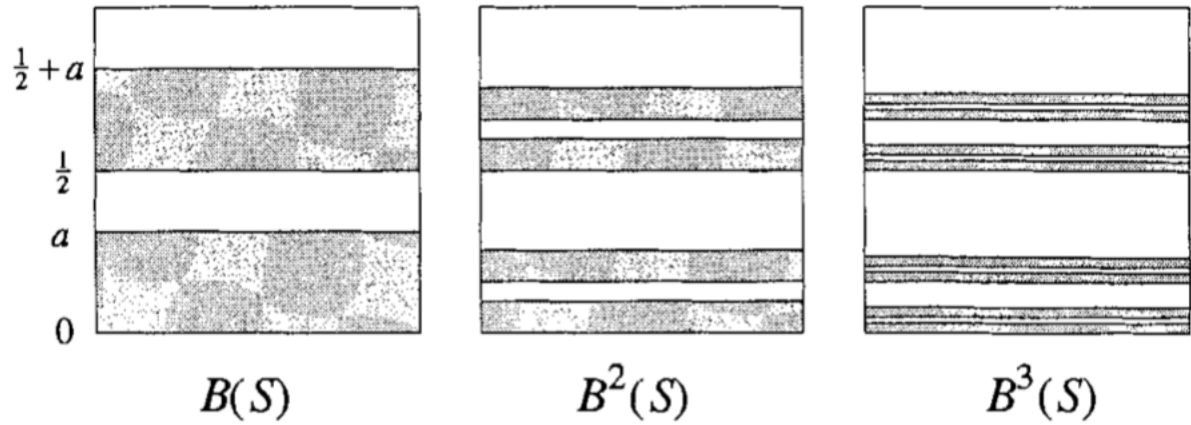


Figure 12.1.5

Hénon map (stretch and fold):

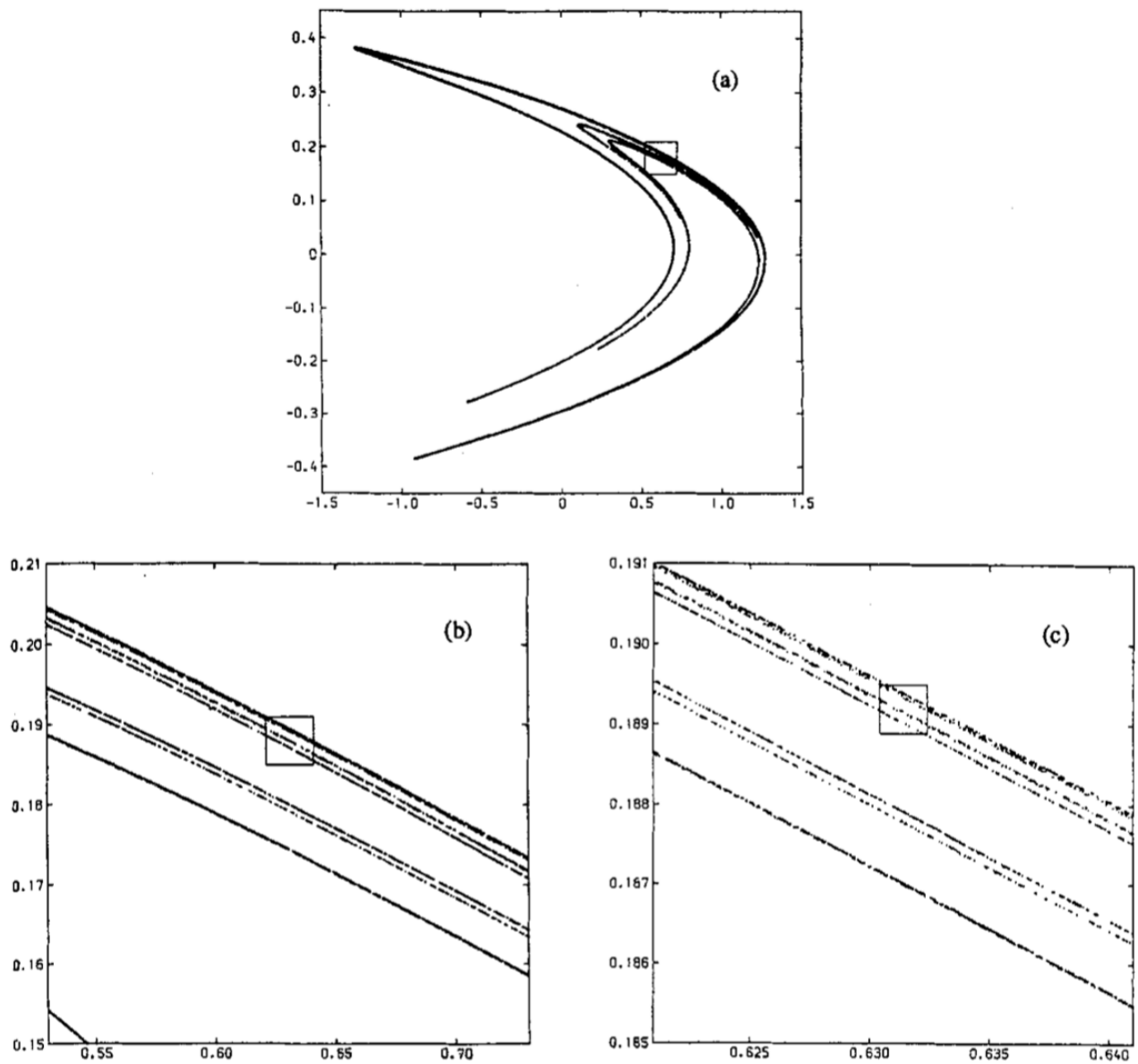


Figure 12.2.3 Hénon (1976), pp 74–76